CHEM 1301 Practice Exam 3-2 — Answer Key

Worked solutions with course-material citations

Every numeric answer was recomputed by `_scripts/build_exam_3_2.py`, which builds this key and the exam from the same data. Atomic masses are periodic-table values (H 1.008, C 12.01, N 14.01, O 16.00, …), matching Ch 8 slides 5, 9, and 22. Citations name the slide (same number in the `.pdf` and `.pptx`) where the content is taught, plus the homework screenshot with the closest problem.

Score bands. 32–35 exam-ready · 27–31 review the missed topics · 26 or below rework the study guide sections cited.

Q	Ans	Answer	Type	Topic (source)
1	**a**	Formation of water	concept	Driving force of a neutralization (Ch 7 sl. 3, 37)
2	**b**	$BaSO_4$	concept	Solubility rules: insoluble salt (Ch 7 sl. 11)
3	**e**	K_2S	concept	Solubility rules: cation rule wins (Ch 7 sl. 11)
4	**a**	$PbSO_4$	concept	Predicting the precipitate (Ch 7 sl. 11, 14)
5	**b**	$Ba^{2+}(aq) + SO_4{}^{2-}(aq) \rightarrow BaSO_4(s)$	concept	Net ionic equation (Ch 7 sl. 17–18)
6	**e**	12	calc	Balancing a neutralization (Ch 7 sl. 27)
7	**e**	K^+ and $NO_3{}^-$	concept	Spectator ions (Ch 7 sl. 15–17)
8	**d**	$KNO_3(aq)$ and $H_2O(l)$	concept	Products of strong acid + strong base (Ch 7 sl. 23, 26)
9	**e**	NaOH and KOH	concept	Identifying strong bases (Ch 7 sl. 25)
10	**b**	separated Ba^{2+} and $NO_3{}^-$ ions, two $NO_3{}^-$ for every Ba^{2+}	concept	Ions in solution (Ch 7 sl. 6–7)
11	**d**	2	calc	Electrons gained per atom (Ch 7 sl. 31, 38)
12	**a**	gas evolution and acid–base	concept	Classifying a carbonate + acid reaction (Ch 7 sl. 39)
13	**d**	decomposition reaction	concept	Recognizing decomposition (Ch 7 sl. 43)

Q	Ans	Answer	Type	Topic (source)
14	**a**	12.01 amu	concept	Atomic mass units (Ch 8 sl. 5, 7)
15	**b**	boron (10.81 g/mol)	concept	Comparing numbers of atoms (Ch 8 sl. 6, 9)
16	**a**	0.448 mol	calc	Grams to moles (Ch 8 sl. 11–12)
17	**d**	1.66×10^{24} atoms	calc	Moles to atoms (Ch 8 sl. 13–14)
18	**c**	0.528 g	calc	Atoms to grams (Ch 8 sl. 15)
19	**e**	310.2 g/mol	calc	Molar mass with parentheses (Ch 8 sl. 22, 26)
20	**d**	34.3 g	calc	Moles to grams (Ch 8 sl. 24, 27)
21	**a**	2.72×10^{-3} mol	calc	Milligrams to moles (Ch 8 sl. 28–29)
22	**c**	21.20%	calc	Mass percent (Ch 8 sl. 30–32)
23	**c**	C_2H_2 and C_6H_6	concept	Same empirical formula (Ch 8 sl. 39)
24	**e**	Fe_2O_3	calc	Empirical formula needing a multiplier (Ch 8 sl. 42–43)
25	**d**	N_2O_4	calc	Empirical to molecular formula (Ch 8 sl. 47–49)
26	**a**	burned in excess oxygen so it fully converts to products, giving the percent C (and other elements) and so the empirical formula	concept	Combustion analysis (Ch 8 sl. 44)
27	**b**	The mole ratios used in the calculation come from the coefficients of the balanced equation	concept	Why balance first (Ch 9 sl. 14, 16)
28	**c**	9	calc	Counting molecules from coefficients (Ch 9 sl. 3–4)
29	**b**	0.429 mol	calc	Mole–mole conversion (Ch 9 sl. 9, 11)
30	**c**	side reactions consume some of the reactants or products	concept	Why actual yield is lower (Ch 9 sl. 28)
31	**d**	38.8 g	calc	Grams to grams (Ch 9 sl. 14, 18)
32	**b**	20.2 g	calc	Limiting reactant, then product mass (Ch 9 sl. 22–23)
33	**c**	23.8 g	calc	Theoretical yield with excess reactant (Ch 9 sl. 25–26)
34	**c**	73.4%	calc	Percent yield with a limiting reactant (Ch 9 sl. 27, 30)

Q	Ans	Answer	Type	Topic (source)
35	**e**	Something is wrong, such as a wet or impure product or a calculation error, because the theoretical yield is the maximum possible	concept	A yield above 100% (Ch 9 sl. 28–29)

Mix: 18 concept / 17 calculation · Ch 7: 13 · Ch 8: 13 · Ch 9: 9

1. Driving force of a neutralization — a) Formation of water

Source: Chapter 7 post, slide 3 (Driving Forces), slide 37 (Acid–Base Reactions: HCl + KOH)
Homework match: HW16 — hw16ex2.png (the same net ionic process, $H^+ + OH^- \rightarrow H_2O$, for any strong acid/strong base)
What it tests: Matching each reaction type to its driving force (acid–base formation of water).

Slide 37 uses this exact reaction: "You can recognize this as an acid–base reaction because it involves an H^+ ion that ends up in the product water." Forming water is driving force #2 on slide 3.

Why the other choices are wrong:

- b) Transfer of electrons: No electrons move. $H^+ + OH^- \rightarrow H_2O$ is acid–base, not redox.
- c) An increase in the number of ions in solution: Not one of the four driving forces. Ions are actually used up.
- d) Formation of a solid: KCl is soluble (rules 2, 3). Nothing precipitates.
- e) Formation of a gas: No gas is produced.

2. Solubility rules: insoluble salt — b) $BaSO_4$

Source: Chapter 7 post, slide 11 (Solubility Rules, Table 7.1), rule 4
Homework match: HW14 — hw14ex2.png (choose the substance not likely to be soluble)
What it tests: The named sulfate exceptions ($BaSO_4$, $PbSO_4$, $CaSO_4$), and rule 2 overriding rule 6.

Rule 4: most sulfates are soluble; **notable exceptions are $BaSO_4$, $PbSO_4$, and $CaSO_4$.** Ca shows up as a *sulfate* exception, not a chloride exception, so $CaCl_2$ is soluble but $CaSO_4$ is not.

Why the other choices are wrong:

- a) K_2CO_3: Rule 2 beats rule 6: K^+ salts are soluble.
- c) NH_4NO_3: Rules 1 and 2: both ions make it soluble.
- d) $CaCl_2$: Rule 3: most chlorides are soluble, and Ca isn't a chloride exception (only Ag, Pb, Hg_2 are).
- e) Na_3PO_4: Rule 2 beats rule 6: Na^+ salts are soluble.

3. Solubility rules: cation rule wins — e) K_2S

Source: Chapter 7 post, slide 11 (Solubility Rules), rules 2 and 6
Homework match: HW14 — hw14ex3.png / hw14ex4.png (which rule explains solubility, e.g. K_3PO_4, ammonium sulfide)
What it tests: Rule 2 (Na^+, K^+, NH_4^+ salts soluble) overrides the 'slightly soluble' anion rule 6.

Sulfides are normally only slightly soluble (rule 6), but **most salts of K^+ are soluble** (rule 2), and the cation rule wins. K_2S is soluble. Every other choice is either a named exception or falls under rule 5 or 6 with a non-rule-2 cation.

Why the other choices are wrong:

- a) $PbSO_4$: Rule 4 exception.
- b) AgCl: Rule 3 exception.
- c) $Ca_3(PO_4)_2$: Rule 6: phosphates are only slightly soluble, and Ca^{2+} isn't a rule-2 cation.
- d) $Fe(OH)_3$: Rule 5: only NaOH and KOH are listed as soluble hydroxides.

4. Predicting the precipitate — a) $PbSO_4$

Source: Chapter 7 post, slide 14 (How to Predict Precipitates), slide 11 (rule 4 exception), slide 10
Homework match: HW15 — hw15ex4.png (molecular equations for precipitation reactions)
What it tests: Swapping anions, then checking both products against the rules, including the exception lists.

The ions present are Pb^{2+}, NO_3^-, Na^+, SO_4^{2-}. Swapping anions gives $PbSO_4$ and $NaNO_3$. $NaNO_3$ is soluble. $PbSO_4$ is a rule-4 exception, so **$PbSO_4$ precipitates**:

$Pb(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow PbSO_4(s) + 2NaNO_3(aq)$

Why the other choices are wrong:

- b) NaPb: Two cations can't form a solid (slide 10).
- c) $NaNO_3$: Rules 1 and 2: soluble.
- d) $Pb(SO_4)_2$: Pb^{2+} and SO_4^{2-} combine 1 : 1. This formula isn't neutral.
- e) No precipitate forms: $PbSO_4$ is a rule-4 exception, so a solid does form.

5. Net ionic equation — b) $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

Source: Chapter 7 post, slides 15–18 (molecular / complete ionic / net ionic equations), slide 11 (rule 4)
Homework match: HW15 — hw15ex1.png (what a net ionic equation shows) and hw15ex2.png (write the net ionic equation)
What it tests: Writing the net ionic equation by cancelling spectator ions, with the precipitate written as (s).

Molecular: $BaCl_2(aq) + K_2SO_4(aq) \rightarrow BaSO_4(s) + 2KCl(aq)$. Write every soluble strong electrolyte as ions. K^+ and Cl^- appear unchanged on both sides, so cancel them: **$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$**.

Why the other choices are wrong:

- a) $K^+(aq) + Cl^-(aq) \rightarrow KCl(s)$: KCl is soluble, so these ions are spectators.
- c) $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(aq)$: $BaSO_4$ precipitates, so it must be (s).

- d) $Ba^{2+}(aq) + 2Cl^{-}(aq) + 2K^{+}(aq) + SO_4{}^{2-}(aq) \rightarrow BaSO_4(s) + 2K^{+}(aq) + 2Cl^{-}(aq)$: Complete ionic equation. It still has the spectators K^+ and Cl^-.
- e) $BaCl_2(aq) + K_2SO_4(aq) \rightarrow BaSO_4(s) + 2KCl(aq)$: Molecular equation.

6. Balancing a neutralization — e) 12

Source: Chapter 7 post, slide 27 (Practice: balancing $H_3PO_4 + Ca(OH)_2 \rightarrow Ca_3(PO_4)_2 + H_2O$)
Homework match: HW22 — balancing with the lowest whole-number coefficients ($2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$)
What it tests: Balancing an acid–base reaction: polyatomic ions first as units, then H and O, then add up the coefficients.

Balance Al: 2 $Al(OH)_3$. Balance SO_4 as a unit: 3 H_2SO_4. H^+ from the acid = 6 and OH^- from the base = 6, so 6 H_2O.

$$3\,H_2SO_4 + 2\,Al(OH)_3 \rightarrow Al_2(SO_4)_3 + 6\,H_2O$$

Sum = 3 + 2 + 1 + 6 = **12**. Check O: 12 + 6 = 18 on the left, 12 + 6 = 18 on the right.

Why the other choices are wrong:

- a) 4: Leaves every coefficient as 1 (unbalanced).
- b) 6: That is the coefficient of H_2O alone.
- c) 8: Balanced Al and SO_4 but left H_2O at 2.
- d) 11: Gives H_2SO_4 a coefficient of 2 instead of 3.

7. Spectator ions — e) K^+ and $NO_3{}^-$

Source: Chapter 7 post, slides 15–17 (molecular → complete ionic → net ionic for $K_2CrO_4 + Ba(NO_3)_2$)
Homework match: HW15 — hw15ex1.png (spectator ions are left out of the net ionic equation)
What it tests: Course definition: spectator ions do not participate directly in the reaction.

Complete ionic: $2K^+ + CrO_4{}^{2-} + Ba^{2+} + 2NO_3{}^- \rightarrow BaCrO_4(s) + 2K^+ + 2NO_3{}^-$. The ions that appear unchanged on both sides are **K^+ and $NO_3{}^-$**. The net ionic equation on slide 17 is $Ba^{2+} + CrO_4{}^{2-} \rightarrow BaCrO_4(s)$.

Why the other choices are wrong:

- a) $NO_3{}^-$ and $CrO_4{}^{2-}$: $CrO_4{}^{2-}$ ends up in the precipitate.
- b) Ba^{2+} and $CrO_4{}^{2-}$: These form the solid, so they are directly involved.
- c) There are no spectator ions: K^+ and $NO_3{}^-$ are unchanged on both sides.
- d) K^+ and Ba^{2+}: Ba^{2+} ends up in the precipitate.

8. Products of strong acid + strong base — d) $KNO_3(aq)$ and $H_2O(l)$

Source: Chapter 7 post, slide 23 (Acid–Base Reactions: HNO_3 + KOH), slide 26 (principle 5: water + a dissolved salt)
Homework match: HW16 — hw16ex2.png (what salt forms: HI + NaOH → NaI, NaOH + HNO_3 → $NaNO_3$)
What it tests: Strong acid + strong base gives water + a salt (base cation + acid anion), and the salt stays dissolved.

$HNO_3(aq)$ + KOH(aq) → **$KNO_3(aq)$ + $H_2O(l)$**. The salt takes K^+ from the base and NO_3^- from the acid. Slide 26, principle 5: the other product "is always an ionic compound called a salt, which remains dissolved in the water."

Why the other choices are wrong:

- a) $KNO_2(aq)$ and $H_2O(l)$: Nitric acid is HNO_3, so the anion is NO_3^-, not NO_2^-.
- b) KH(aq) and $NO_3OH(aq)$: The acid's H^+ joins OH^- to make water. These formulas aren't real.
- c) $K_2O(s)$ and $HNO_3(aq)$: Acid + base makes a salt + water.
- e) $KNO_3(s)$ and $H_2O(l)$: KNO_3 is soluble (rules 1, 2). The salt stays dissolved (slide 26, principle 5).

9. Identifying strong bases — e) NaOH and KOH

Source: Chapter 7 post, slide 25 (principle 3: 'The most common strong bases are NaOH and KOH'), slide 11 (rule 5)
Homework match: HW16 — hw16ex1.png (what is meant by a strong base)
What it tests: The course's strong-base list, and the link to solubility rule 5 (NaOH and KOH are the soluble hydroxides).

Slide 25: a strong base is a metal hydroxide that is very soluble in water, and "the most common strong bases are **NaOH and KOH**." These are also the two hydroxides rule 5 names as soluble, which is no coincidence. A hydroxide that barely dissolves can't release much OH^-.

Why the other choices are wrong:

- a) HCl and NaOH: HCl is a strong *acid*.
- b) $Mg(OH)_2$ and $Fe(OH)_3$: Only slightly soluble hydroxides (rule 5), so they aren't strong bases.
- c) KOH and HNO_3: HNO_3 is a strong *acid*.
- d) $Ca(OH)_2$ and $Al(OH)_3$: $Ca(OH)_2$ is only moderately soluble and $Al(OH)_3$ is only slightly soluble.

10. Ions in solution — b) separated Ba^{2+} and NO_3^- ions, two NO_3^- for every Ba^{2+}

Source: Chapter 7 post, slide 6 (Ionic Compounds in Water: the $Ba(NO_3)_2$ example), slide 7 (Strong Electrolytes)
Homework match: HW15 — complete vs net ionic equations (strong electrolytes written as ions)
What it tests: Soluble ionic compounds are strong electrolytes that break into separated ions in the formula ratio.

Slide 6: "$Ba(NO_3)_2(aq)$ does not contain $Ba(NO_3)_2$ units. Rather, it contains separated Ba^{2+} and NO_3^- ions … two NO_3^- ions for every Ba^{2+} ion." The ions are why the solution conducts electricity (slide 7).

Why the other choices are wrong:

- a) Ba^{2+}, N^{3-}, and O^{2-} ions: Polyatomic ions such as NO_3^- stay together.
- c) separated Ba^{2+} and NO_3^- ions, one NO_3^- for every Ba^{2+}: Charge balance needs two NO_3^- per Ba^{2+}.
- d) neutral Ba atoms and NO_3 molecules: Dissolving an ionic solid releases ions, not neutral atoms. That's why the solution conducts.
- e) intact $Ba(NO_3)_2$ units: Slide 6: $Ba(NO_3)_2(aq)$ does *not* contain $Ba(NO_3)_2$ units.

11. Electrons gained per atom — d) 2

Source: Chapter 7 post, slide 31 (metal + nonmetal: the nonmetal atom becomes an anion), slide 38 (2Li + F_2: each F atom gains one electron), slide 33
Homework match: HW16 — hw16ex3.png (Ba + Cl_2: each Ba atom loses two electrons)
What it tests: Electrons gained per nonmetal atom, read from the charge of its ion in the ionic product.

Step 1: find the oxygen ion in the product. Al_2O_3 is ionic (metal + nonmetal, slide 31), so it is made of Al^{3+} and O^{2-} ions. Check that the charges cancel: $2(+3) + 3(-2) = 0$.

Step 2: read the electrons from the charge. Each O atom starts neutral (charge 0) in O_2 and ends as an O^{2-} ion. Going from 0 to −2 means it **gains 2 electrons**: $O + 2e^- \rightarrow O^{2-}$. This is the same per-atom count as slide 38, where each F atom gains one electron to become F^-.

Check: each Al atom goes from 0 to +3 and loses 3 electrons. In the equation, 4 Al × 3 = 12 lost and 6 O atoms × 2 = 12 gained, so electrons lost = electrons gained.

Why the other choices are wrong:

- a) 6: That is the electrons transferred in one Al_2O_3 formula unit (3 O atoms × 2), not per O atom.
- b) 4: That counts both O atoms from one O_2 molecule. The question asks about one O atom.
- c) 3: That is the electrons **lost** by each Al atom ($Al \rightarrow Al^{3+}$), not gained by O.
- e) 1: That would make O^-. In Al_2O_3, oxygen is O^{2-} (2 Al^{3+} balance 3 O^{2-}).

12. Classifying a carbonate + acid reaction — a) gas evolution and acid–base

Source: Chapter 7 post, slide 39 (Gas Evolution Reactions: the same equation, 'also an acid–base reaction'), slide 45
Homework match: HW17 — hw17ex1.png (classify each reaction as precipitation, acid–base, and/or oxidation–reduction)
What it tests: Gas evolution as a class, and a reaction belonging to two classes at once.

Slide 39 uses this exact equation: "In this reaction carbon dioxide gas is formed. This reaction is also an acid–base reaction." The H^+ from HCl ends up in H_2O, and $CO_2(g)$ forms, so it is both **gas evolution and acid–base**.

Why the other choices are wrong:

- b) precipitation only: No solid forms; NaCl is soluble.
- c) combustion: No O_2 and no flame.

- d) decomposition: Two reactants combine. This isn't one compound breaking apart.
- e) oxidation–reduction and single-replacement: No electrons are transferred; every ion keeps its charge.

13. Recognizing decomposition — d) decomposition reaction

Source: Chapter 7 post, slide 43 (Decomposition: a compound broken into simpler substances, usually by heat or current), slide 42
Homework match: HW17 — decomposition reactions (ammonium nitrite → N_2 + H_2O; HBr → H_2 + Br_2)
What it tests: Decomposition: one compound breaks into simpler compounds or elements, usually by heating.

One compound, $KClO_3$, breaks into simpler substances (KCl and O_2) when heated. That matches slide 43's definition of **decomposition** (compare 2HgO → 2Hg + O_2 on the same slide). Ch 9 slide 32 uses this same reaction for percent yield.

Why the other choices are wrong:

- a) synthesis reaction: Synthesis makes one compound from simpler materials. This is the reverse.
- b) acid–base reaction: No H^+ or OH^- is involved.
- c) double-displacement reaction: There is only one reactant, so there are no anions to swap.
- e) combustion reaction: O_2 is a product here, not a reactant burning something.

14. Atomic mass units — a) 12.01 amu

Source: Chapter 8 post, slide 5 (Atomic Mass Units, Table 8.1: C = 12.01 amu), slide 7 (The Mole: 12.01 g of C)
Homework match: HW18 — 'use the average atomic masses' conversions
What it tests: Average atomic mass in amu (one atom) versus molar mass in g (one mole).

Table 8.1 gives carbon's average atomic mass as **12.01 amu** for one atom. One *mole* of C atoms has a mass of 12.01 g (slide 7). The same number, but different units and different amounts.

Why the other choices are wrong:

- b) 1 amu: 1 amu is about the mass of one H atom (1.008 amu).
- c) 6.022×10^{23} g: That is Avogadro's *number*, not a mass.
- d) 1.66×10^{-24} amu: 1 amu = 1.66×10^{-24} **g**. The units are mixed up.
- e) 12.01 g: 12.01 g is the mass of one *mole* of C atoms.

15. Comparing numbers of atoms — b) boron (10.81 g/mol)

Source: Chapter 8 post, slide 9 (Table 8.2: 1-mole samples of Al, Au, Fe, S, B, Xe), slide 6, slide 2 (counting by weighing)
Homework match: HW18 — comparing samples (which contains the greater number of atoms?)
What it tests: For equal masses, moles = mass / molar mass, so the lightest atoms give the most atoms.

Moles = 10.0 g ÷ molar mass, and atoms = moles × N_A. The smallest molar mass gives the most moles and so the most atoms: B: 10.0 / 10.81 = 0.925 mol, compared with Au: 10.0 / 196.97 = 0.0508 mol. **Boron** wins.

Why the other choices are wrong:

- a) gold (196.97 g/mol): Heaviest atoms here, so the *fewest* atoms.
- c) aluminum (26.98 g/mol): Heavier atoms, so fewer of them in 10.0 g.
- d) iron (55.85 g/mol): Heavier atoms, so fewer of them.
- e) xenon (131.3 g/mol): Heavier atoms, so fewer of them.

16. Grams to moles — a) 0.448 mol

Source: Chapter 8 post, slides 11–12 (Mole Conversion Practice: 3.60 g Al, 544 g Ca)
Homework match: HW18 — hw18ex1.png (moles of element from grams); 45.9 g Ge → mol
What it tests: Grams → moles: divide by the molar mass.

$$25.0 \text{ g Fe} \times \frac{1 \text{ mol}}{55.85 \text{ g}} = 0.448 \text{ mol Fe}$$

Why the other choices are wrong:

- b) 2.23 mol: Inverted: 55.85 / 25.0.
- c) 4.48 mol: Decimal slip (off by 10).
- d) 1396 mol: Multiplied by the molar mass.
- e) 2.70×10^{23} mol: That is the number of atoms.

17. Moles to atoms — d) 1.66×10^{24} atoms

Source: Chapter 8 post, slides 13–14 (Atom Conversion Practice: 0.133 mol Al, 13.6 mol Ca), slide 7
Homework match: HW18 — hw18ex3.png (number of atoms from 0.00580 mol Cu)
What it tests: Moles → atoms uses only Avogadro's number. The molar mass isn't needed.

$$2.75 \text{ mol} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}} = 1.66 \times 10^{24} \text{ atoms Ca}$$

Why the other choices are wrong:

- a) 4.57×10^{-24} atoms: Divided by Avogadro's number instead of multiplying.
- b) 4.13×10^{22} atoms: Treated 2.75 mol as 2.75 g and divided by the molar mass first.
- c) 1.66×10^{23} atoms: Exponent slip.
- e) 6.64×10^{25} atoms: Multiplied by the molar mass too. You don't need it for mol → atoms.

18. Atoms to grams — c) 0.528 g

Source: Chapter 8 post, slide 15 (Convert atoms to grams: 2.31×10^{22} atoms of gold)
Homework match: HW18 — atoms → grams setup ((1 mol / 6.022×10^{23} atoms)(g / 1 mol))
What it tests: Atoms → moles ($\div$ N_A) → grams ($\times$ molar mass).

$$5.00 \times 10^{21} \text{ atoms} \times \frac{1 \text{ mol}}{6.022 \times 10^{23}} = 8.30 \times 10^{-3} \text{ mol}; \quad 8.30 \times 10^{-3} \times 63.55 = 0.528 \text{ g Cu}$$

Why the other choices are wrong:

- a) 1.31×10^{-4} g: Divided by the molar mass instead of multiplying.
- b) 8.30×10^{-3} g: Stopped at moles.
- d) 7.65×10^{3} g: Inverted: 63.55 / mol.
- e) 3.18×10^{23} g: Never divided by Avogadro's number.

19. Molar mass with parentheses — e) 310.2 g/mol

Source: Chapter 8 post, slide 22 (Molar Mass), slide 26 (LA problem: $Ba(OH)_2$, parentheses)
Homework match: HW19 — hw19ex1.png (formula and molar mass)
What it tests: Counting atoms when a subscript multiplies a polyatomic ion.

Atom count: 3 Ca, 2 P, 8 O.

$$3(40.08) + 2(30.97) + 8(16.00) = 120.24 + 61.94 + 128.00 = 310.18 \approx 310.2 \text{ g/mol}$$

Why the other choices are wrong:

- a) 135.1 g/mol: Ignored every subscript outside the parentheses.
- b) 215.2 g/mol: Dropped the 2 outside the parentheses.
- c) 230.0 g/mol: Dropped the 3 on Ca.
- d) 246.2 g/mol: Applied the 2 to P but not to O.

20. Moles to grams — d) 34.3 g

Source: Chapter 8 post, slide 24 (molar mass of H_2SO_4), slide 27 (Calculating Mass from Moles)
Homework match: HW18 — hw18ex2.png (mass in grams from moles)
What it tests: Moles → grams: multiply by the molar mass.

Molar mass $H_2SO_4 = 2(1.008) + 32.07 + 4(16.00) = 98.09$ g/mol.

$$0.350 \text{ mol} \times 98.09 \ \tfrac{\text{g}}{\text{mol}} = 34.3 \text{ g}$$

Why the other choices are wrong:

- a) 3.57×10^{-3} g: Divided by the molar mass.
- b) 11.2 g: Used only the mass of S.
- c) 28.7 g: Wrong formula: H_2SO_3 (sulfurous acid) has 3 O.
- e) 280 g: Divided the molar mass by the moles.

21. Milligrams to moles — a) 2.72×10^{-3} mol

Source: Chapter 8 post, slide 29 (LA problem 1: 9.43 mg of calcium chloride), slide 28
Homework match: HW19 — hw19ex2.png (moles in 21.9 mg of nitrogen dioxide)
What it tests: Convert mg → g before dividing by the molar mass, and write the formula correctly from the name.

NO_2: 14.01 + 2(16.00) = 46.01 g/mol. 125 mg = 0.125 g.

$$\frac{0.125 \text{ g}}{46.01 \text{ g/mol}} = 2.72 \times 10^{-3} \text{ mol}$$

Why the other choices are wrong:

- b) 4.17×10^{-3} mol: Used the molar mass of NO (30.01) instead of NO_2.
- c) 2.72 mol: Didn't convert mg to g.
- d) 5.75 mol: Multiplied by the molar mass.
- e) 1.64×10^{21} mol: That is the number of molecules.

22. Mass percent — c) 21.20%

Source: Chapter 8 post, slide 30 (Mass Percent), slide 31 (ethanol example), slide 32 (H_3PO_4 practice)
Homework match: HW20 — hw20ex2.png (percent by mass of the first element listed, e.g. KNO_2, theobromine)
What it tests: Mass percent with a subscript outside the parentheses (2 N atoms).

Molar mass: 2(14.01) + 8(1.008) + 32.07 + 4(16.00) = 132.15 g/mol. Mass of N in 1 mol = 2(14.01) = 28.02 g.

$$\%\text{N} = \frac{28.02}{132.15} \times 100\% = 21.20\%$$

Why the other choices are wrong:

- a) 10.60%: Counted only one N; the formula has 2.
- b) 13.33%: Atom fraction (2 N of 15 atoms), not mass.
- d) 24.27%: That is % S.
- e) 48.43%: That is % O.

23. Same empirical formula — c) C_2H_2 and C_6H_6

Source: Chapter 8 post, slide 39 (Empirical and Molecular Formulas: $C_6H_{12}O_6$ and CH_2O-type compounds share an empirical formula), slide 40
Homework match: HW21 — CH has a molar mass of about 26, so C_2H_2
What it tests: Reducing molecular formulas to the smallest whole-number ratio and comparing them.

$C_2H_2 \div 2 = \mathbf{CH}$, and $C_6H_6 \div 6 = \mathbf{CH}$. Both have the empirical formula CH. They are different compounds (different molar masses) with the same ratio of atoms, which is slide 39's point.

Why the other choices are wrong:

- a) CO and CO_2: Empirical formulas CO and CO_2 differ.
- b) C_2H_4 and C_2H_6: Empirical formulas CH_2 and CH_3 differ.
- d) H_2O and H_2O_2: Empirical formulas H_2O and HO differ.
- e) NO and NO_2: Empirical formulas NO and NO_2 differ.

24. Empirical formula needing a multiplier — e) Fe_2O_3

Source: Chapter 8 post, slides 42–43 (Steps for Determining the Empirical Formula, step 4: multiply by the smallest integer), slides 35–37
Homework match: HW21 — empirical formula from percent composition
What it tests: Step 4 of the empirical-formula method: clearing a .5 ratio by multiplying by 2.

Per 100 g: Fe 69.94 / 55.85 = 1.252 mol and O 30.06 / 16.00 = 1.879 mol. Divide by the smallest: Fe 1.00, O 1.50. 1.50 isn't a whole number, so multiply both by 2: Fe 2, O 3. The formula is $\mathbf{Fe_2O_3}$.

Why the other choices are wrong:

- a) FeO_2: Rounded 1.5 up to 2.
- b) Fe_3O_4: Mole ratio 1 : 1.33. That doesn't match 1 : 1.5.
- c) FeO: Rounded 1.5 down to 1. Multiply instead (slide 43, step 4).
- d) Fe_3O_2: Inverted ratio; there is more O than Fe in moles.

25. Empirical to molecular formula — d) N_2O_4

Source: Chapter 8 post, slide 48 ($(\text{empirical formula})_n$ = molecular formula), slide 49 (LA Practice), slide 47
Homework match: HW21 — molecular formula from empirical formula + molar mass
What it tests: n = molar mass / empirical mass, and every subscript gets multiplied by n.

Empirical mass NO_2 = 14.01 + 2(16.00) = 46.01. n = 92.02 / 46.01 = 2, so $(NO_2)_2 = \mathbf{N_2O_4}$.

Why the other choices are wrong:

- a) NO_4: Multiplied only the O subscript by 2.
- b) NO_2: n = 1 would give a molar mass of 46.01.
- c) N_4O_8: n = 4 would give 184.
- e) N_3O_6: n = 3 would give 138.

26. Combustion analysis — a) burned in excess oxygen so it fully converts to products, giving the percent C (and other elements) and so the empirical formula

Source: Chapter 8 post, slide 44 (Combustion analysis), slide 47
Homework match: HW21 — hw21ex1.png (what evidence is needed for empirical vs molecular formulas)
What it tests: What combustion analysis does and what it produces (composition, leading to the empirical formula).

Slide 44: combustion analysis "works by burning the sample in excess oxygen so that it fully converts to products. … we can find the percent of carbon and the other elements which leads us to the empirical formula." To go further to the molecular formula, you would still need the molar mass (slide 47).

Why the other choices are wrong:

- b) heated with an electric current until it decomposes into its elements: That describes a decomposition reaction (slide 43).
- c) weighed to determine its molecular formula without any other data: Mass alone gives no formula. The molecular formula needs composition plus molar mass.
- d) burned in a limited amount of oxygen to measure its molar mass directly: Combustion analysis uses *excess* O_2 and gives composition, not molar mass.
- e) dissolved in water to see whether it conducts electricity: That tests for electrolytes (slide 7).

27. Why balance first — b) The mole ratios used in the calculation come from the coefficients of the balanced equation

Source: Chapter 9 post, slide 16 (Think, Pair, Share 9.3: why balance before stoichiometry?), slide 14 (step 1: balance), slide 2
Homework match: HW22 — balancing before mole-ratio problems ('for each of the following unbalanced equations …')
What it tests: The reason behind step 1 of slide 14: coefficients give the mole ratios.

Slide 2: "The coefficients in a balanced equation give us the relative numbers of molecules." Every stoichiometry problem uses a mole ratio (slide 14, step 3). With wrong coefficients, every ratio, and every answer, is wrong.

Why the other choices are wrong:

- a) Balancing is needed only so the molar masses can be calculated: Molar masses come from the formulas, not the coefficients.
- c) Balancing converts the coefficients from grams into moles: Coefficients are always mole/particle ratios. They are never grams.
- d) Balancing makes the total mass of the reactants equal the mass of each product: Mass is conserved in total, not per substance.
- e) An unbalanced equation has no limiting reactant: Limiting reactants exist either way. You just can't find them without the correct ratio.

28. Counting molecules from coefficients — c) 9

Source: Chapter 9 post, slide 4 (Chemical Equation Practice: $2Al_2O_3 \rightarrow 4Al + 3O_2$), slide 3 (Table 9.1)
Homework match: HW22 — interpreting a balanced equation in molecules and moles ($H_2 + F_2 \rightarrow 2HF$)
What it tests: Coefficients as particle ratios, scaled to a given number of particles.

2 $Al_2O_3 \Leftrightarrow$ 3 O_2, so 6 $Al_2O_3 \times$ (3 O_2 / 2 Al_2O_3) = **9 O_2 molecules** (and 12 Al atoms).

Why the other choices are wrong:

- a) 3: That is the coefficient, not scaled to 6 units of Al_2O_3.
- b) 4: Inverted the ratio: $6 \times 2/3$.
- d) 12: That is the number of Al atoms produced.

- e) 18: Multiplied by 3 without dividing by 2.

29. Mole–mole conversion — b) 0.429 mol

Source: Chapter 9 post, slide 11 (Mole–Mole Conversions Practice 2: $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$), slide 9 (ethane combustion)
Homework match: HW22 — moles of product from moles (e.g. $KO_2 + H_2O$: mol O_2 from 0.480 mol KOH)
What it tests: Picking the right two coefficients from a four-substance equation.

$$0.750 \text{ mol } O_2 \times \frac{4 \text{ mol } CO_2}{7 \text{ mol } O_2} = 0.429 \text{ mol } CO_2$$

Why the other choices are wrong:

- a) 0.214 mol: Used the C_2H_6 coefficient (2) instead of CO_2 (4).
- c) 0.750 mol: Assumed 1 : 1.
- d) 1.31 mol: Inverted ratio (7/4).
- e) 1.50 mol: Multiplied by 2.

30. Why actual yield is lower — c) side reactions consume some of the reactants or products

Source: Chapter 9 post, slide 28 (Actual Yield: 'One reason for this is the presence of side reactions'), slide 25
Homework match: HW24 — percent yield problems (actual is less than theoretical)
What it tests: Course definition of actual yield and the reason it falls short.

Slide 28: "the amount of product predicted (the theoretical yield) is seldom obtained. One reason for this is the presence of **side reactions** (other reactions that consume one or more of the reactants or products)."

Why the other choices are wrong:

- a) mass is not conserved in most chemical reactions: Mass is always conserved.
- b) the excess reactant also turns into product: The excess reactant is left over. It doesn't add product.
- d) the limiting reactant is never used up: By definition, the limiting reactant runs out first.
- e) the theoretical yield is calculated from the excess reactant: The theoretical yield comes from the *limiting* reactant (slide 25).

31. Grams to grams — d) 38.8 g

Source: Chapter 9 post, slide 18 (Stoichiometry Practice 2: $H_2SO_4 + 2KOH$), slide 14 (steps)
Homework match: HW23 — mass of product from mass of reactant ($CaC_2 \rightarrow C_2H_2$)
What it tests: g → mol → mol ratio → g, with a 2 : 1 ratio.

$$25.0 \text{ g KOH} \div 56.11 = 0.4456 \text{ mol}; \quad \times\tfrac{1}{2} = 0.2228 \text{ mol } K_2SO_4; \quad \times 174.27 = 38.8 \text{ g}$$

Why the other choices are wrong:

- a) 0.223 g: That is the moles of K_2SO_4.
- b) 21.9 g: Used the molar mass of H_2SO_4 instead of K_2SO_4.
- c) 25.0 g: Assumed the product mass equals the reactant mass.
- e) 77.6 g: Assumed 1 : 1 (forgot the 2 KOH).

32. Limiting reactant, then product mass — b) 20.2 g

Source: Chapter 9 post, slide 23 (Limiting Reactants Practice: $3Mg + N_2 \rightarrow Mg_3N_2$), slides 21–22, 25
Homework match: HW23 — limiting reactant then product mass (5.65 g of each reactant; $Fe + O_2$)
What it tests: Find the limiting reactant from moles, then calculate the product from it.

Mg needs 0.600 / 3 = 0.200 mol N_2. 0.250 mol is available, so N_2 is in excess and **Mg is limiting**.

$$0.600 \text{ mol Mg} \times \frac{1 \text{ mol } Mg_3N_2}{3 \text{ mol Mg}} \times 100.95 \ \tfrac{\text{g}}{\text{mol}} = 20.2 \text{ g}$$

Why the other choices are wrong:

- a) 0.200 g: That is the moles of Mg_3N_2.
- c) 21.6 g: Added the reactant masses, ignoring the excess N_2.
- d) 25.2 g: Treated N_2 as limiting. It is in excess (Mg needs only 0.200 mol N_2).
- e) 60.6 g: Assumed 1 mol Mg gives 1 mol Mg_3N_2.

33. Theoretical yield with excess reactant — c) 23.8 g

Source: Chapter 9 post, slide 26 (Theoretical Yield Practice: 237 g NaCl + excess $Pb(NO_3)_2$), slide 25
Homework match: HW24 — theoretical yield calculations (XeF_4; $Na_2S_2O_3 \cdot 5H_2O$)
What it tests: 'Excess' tells you which reactant is limiting, so calculate the yield from it.

$$10.0 \text{ g NaCl} \div 58.44 = 0.1711 \text{ mol}; \quad \times\tfrac{1}{2} = 0.08556 \text{ mol } PbCl_2; \quad \times 278.1 = 23.8 \text{ g}$$

Why the other choices are wrong:

- a) 0.0856 g: That is the moles of $PbCl_2$.
- b) 10.0 g: Assumed the product mass equals the reactant mass.
- d) 28.3 g: Used the molar mass of $Pb(NO_3)_2$.
- e) 47.6 g: Assumed 1 : 1 (forgot the 2 NaCl).

34. Percent yield with a limiting reactant — c) 73.4%

Source: Chapter 9 post, slide 27 (theoretical yield with two given masses), slide 30 (Percent Yield formula), slides 31–32
Homework match: HW24 — hw24ex2.png (theoretical yield of XeF_4 from two masses, then percent yield)
What it tests: The full chain: limiting reactant → theoretical yield → percent yield.

N_2: 14.0 / 28.02 = 0.4996 mol. H_2: 6.00 / 2.016 = 2.976 mol. N_2 needs 3 × 0.4996 = 1.499 mol H_2, which is less than 2.976, so **N_2 is limiting**.

$$NH_3 \text{ theoretical} = 0.4996 \times 2 \times 17.03 = 17.02 \text{ g}; \qquad \% \text{ yield} = \frac{12.5}{17.02} \times 100 = 73.4\%$$

Why the other choices are wrong:

- a) 37.0%: Used H_2 as limiting. It is in excess.
- b) 62.5%: Divided by the total reactant mass.
- d) 89.3%: Divided by the mass of N_2.
- e) 136%: Upside down: theoretical / actual.

35. A yield above 100% — e) Something is wrong, such as a wet or impure product or a calculation error, because the theoretical yield is the maximum possible

Source: Chapter 9 post, slide 29 (Think, Pair, Share 9.6: can the actual yield exceed the theoretical?), slide 28 (theoretical yield = maximum)
Homework match: HW24 — percent yield
What it tests: The theoretical yield is a ceiling. Percent yield can't truly exceed 100%.

Slide 28 defines the theoretical yield as "the **maximum** amount of product that can be produced." A measured result over 100% means extra mass (water or impurities in the product) or a mistake. It can't be real product.

Why the other choices are wrong:

- a) The theoretical yield must have been calculated from the limiting reactant: That is the correct way to calculate it. It doesn't explain the error.
- b) The excess reactant was converted into additional product: The theoretical yield already accounts for the limiting reactant. The excess can't react further.
- c) Yields above 100% are normal whenever the reaction is fast: The theoretical yield is the maximum, regardless of speed.
- d) Side reactions produced extra product beyond the theoretical yield: Side reactions *lower* the yield (slide 28).